

Alex — Tier 1 sequential model outline

Role of this document

This note is the **advisor-ordered spine**: each section follows the sequence Alex emphasized (minimal model → assumptions → data → fitting → what we extract). It **does not replace** existing work; it points to it.

Stable references (do not fork their content here):

- `Tier1_Briefing_Outline.md` — full technical detail, equations, and data column map.
- `Tier1_Narrative_Outline.md` — narrative arc and voice.
- `sports/documents/tier_1_roadmap.md` — execution contract for `535` / panel columns.
- `sports/538_alex_tier1_model_and_fit.ipynb` — notebook that should **mirror this section order** for code + outputs (`537` stays the simulation lab).
- **Wang-style cross-domain template** (same *logical* steps as this outline): `Vector_Questions_and_Modeling_Thoughts.md` §11; mechanics of the Yin–Wang failure/success paper: `wang_paper_model.md`.

1. Unit of modeling

- State the primary unit (i, j, t) and the empirical row (player–season / player–team–season).
- One sentence: local pool vs global advancement.

Fill from `Tier1_Briefing_Outline.md` §1; keep only what you will say aloud.

2. Minimal model objects (headline only)

- Outcome Y (global advancement).
- Amalgamated local environment L_{ijt} (Tier 1 object); first proxy Q .
- Global selection capacity Λ_t .
- Role of own performance / ability proxy A_{ijt} (baseline, not the mechanism headline).

Detail table: `Tier1_Briefing_Outline.md` §2–§3.

3. Minimal assumptions

- Local vs global scarcity framing.
- Leave-self-out for peer-based Q .

- Variable domains (binary Y , standardized L for fitting, etc.).
- What Tier 1 **defers** (sorting, dynamics, full causal ID).

Bullets only here; expand in manuscript later.

4. Map model to data

One table: object → meaning → `df` column / construction step (`530` / `535` CELL 2–3).

Canonical table: `Tier1_Briefing_Outline.md` §4.

5. Primary estimand

- **Main object to report:** turning point L^* (or Q^* in raw units), not only linear coefficients.
- One sentence on why (competing margins / inverted-U readout).

Formulas: `Tier1_Briefing_Outline.md` §7.

6. Fitting plan (ordered)

1. Descriptive bins (no model).
2. Transparent quadratic spec in L (LPM) + controls as needed.
3. Binary response (logit/probit) for the same index structure.
4. Robustness: FE, clustering, alternative L proxy, one decomposition term at a time.

Expand: `Tier1_Briefing_Outline.md` §5–§6.

7. Where simulation fits (`537`)

- One short subsection: `537` **illustrates** mechanism and comparative statics; it does **not** replace the empirical ladder unless you explicitly bridge assumptions.
 - No requirement to change `537` when this outline or `538` evolve.
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8. Deliverables for the next conversation

- This outline filled in at **talking-point** depth (sub-bullets, not dissertation length).

- Matching cells in `538_alex_tier1_model_and_fit.ipynb` so figures/tables follow sections 4–6.

9. Wang-style program — same skeleton, many domains

Alex’s sequence is how you **lock a minimal empirical model** in one domain. The Wang-line aspiration in your notes is: **(i)** a stable empirical shape, **(ii)** a **small** generative story, **(iii)** extra **testable** predictions, **(iv)** replay in other domains.

Keep constant across domains (the “minimal model that does the trick”):

Layer	What stays the same	What changes per domain
Unit	Individual (or role-holder) embedded in a local pool at time t ; outcome is globally scarce.	Definition of pool (team, unit, department) and calendar.
Objects	Y , amalgamated L (peer/rivalry proxy), Λ_t , baseline A .	How you measure L , Y , Λ from logs.
Estimand	Turning point L^* (or equivalent in z-score then mapped back).	Binning / windowing rules; institutional noise.
Fitting ladder	Bins $\rightarrow$ quadratic in L (LPM) $\rightarrow$ logit/probit $\rightarrow$ one decomposition at a time.	FE structure (season vs fiscal year vs cohort).

How you *find* minimality (not wish it):

1. **Start with the briefing’s Tier 1 spec** — one L proxy, $L + L^2 + A$, report L^* . That *is* the candidate minimal reduced form.
2. **Add complexity only on forensic rules:** e.g. crowding or minutes enters only if (a) theory says the shape should move with that margin *and* (b) the data move that way, or the minimal spec is badly misspecified in a pre-registered sense. Your “one diagnostic at a time” rule is the operational version of minimality.
3. Use `537` for Wang-style *comparative statics* without bloating the empirical notebook: e.g. shift effective Λ , concentration, or opportunity; check whether L^* / top-bin behavior moves in the direction `Vector_Questions` lists (shift of peak, stronger top decline under congestion). That ties “simple mechanism” to **extra predictions**, not only the inverted-U picture.
4. **Per new domain:** copy the **same section headings** (this doc §1–§6) into a short domain addendum or notebook; replace §4’s table with that domain’s column mapping; rerun the ladder. The **abstract** story should read almost unchanged; only the measurement bridge changes.

Honest scope: “All domains” is **parallel replay of the same checklist**, not one pooled mega-regression on day one. The Wang papers earn universality by showing the **same qualitative mechanism** with domain-specific codings; your Alex outline is the contract for what must match across those codings.

Changelog

Date	Note
2026-05-12	§9: Wang-style cross-domain program — same skeleton, measurement bridge per domain; 537 for comparative statics.
2026-05-12	Initial skeleton: Alex-sequenced spine; links to existing briefing + new 538 notebook.